

Round-20: the Characterization Theorem, Proved

`uSource_eq_frame` is now a kernel-checked theorem; `formal/Round19Transport.lean`
builds with zero `sorry`

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July 13, 2026

Abstract

Round-19 left exactly one `sorry`: the characterization obligation `uSource_eq_frame` — on well-formed certificates, the source-oriented update U1–U5 agrees with the operationally unfolded frame on every distinct existing non-core-core pair. That statement, whose prose renditions produced defects #11 and #12, is now a kernel-checked theorem: `formal/Round19Transport.lean` builds with **zero `sorry`**, no axioms beyond Lean’s kernel, and no `native_decide` anywhere in the main proof chain (it remains only in the standalone witness examples). The companion fuel-adequacy obligation is discharged in the same file (`uSourceFuel_irrel`: the update is constant in the fuel above the explicit threshold $2 \cdot \text{maxBirth} + 1$, plus one on flipped orientation), and non-vacuity is witnessed (`certC_wellformed` plus an end-to-end `example`). The mathematically substantive outcome is that the proof *forced four additional wellformedness clauses*, each semantically intended since round 15 and each silently honored by the WP34 generator, none previously stated. Status: *strongly supported, not certified*.

1 What was proved

- `frame_char`: for every wellformed certificate and every $n \leq$ the number of steps, after n canonical steps the frame value of every distinct existing non-core-core pair equals `uSource`. Plain induction on the canonical step list — the induction the thirteenth review’s canonical-order determinism made available. Four cases per step: both endpoints old (first-match monotonicity), old member/fresh (pattern value; the birth flip is `net_conv`), old steered/fresh (steering value; the recursive row reconstruction matches the recorded rows by the induction hypothesis plus fuel irrelevance), co-birth (pattern value — defect #12’s geometry, now one induction case).
- `uSourceFuel_irrel`: round-19’s stated fuel-adequacy obligation. The recursion measure is made exact: $\text{fuelNeed} = 2 \cdot \text{maxBirth} + 1$, plus one on the younger-first orientation; above it the fuel is irrelevant.
- `certC_wellformed` and an `example` applying the main theorem to a concrete certificate with an inherited slot: `Wellformed` is not vacuous, and the theorem fires end to end.
- Supporting layer: a first-match calculus for frame lookup (membership plus value-uniqueness replaces all ordering reasoning), a domain invariant (frame keys mention only existing occurrences), and injectivity of the port map on member ports — the usable form of the twelfth review’s 17.4 discipline.

*Round-20 of the $\mathcal{ALCZ}_{\text{RCC5}}$ decidability project. Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample verdict. Round-19 (the transport layer authored in Lean) is round-20’s substrate; this round discharges its single stated `sorry`.

2 The four forced wellformedness clauses

The proof would not close without four clauses round-19’s “minimal” **Wellformed** lacked. Each is a statement the manuscripts always intended and the WP34 generator always satisfied; the medium made their absence a type error rather than a review finding:

<code>targets_exist</code>	Slot targets are <i>real</i> occurrences (in-range fresh indices). Without it, ghost occurrences enter patterns and pollute recorded rows.
<code>targets_length</code>	Exactly one target per template slot. Without it, out-of-range slot ports alias core 0 under the port map’s default and injectivity fails.
<code>net_conv</code>	Pattern nets are converse-coherent — the R2 half of “patterns are closed atomic networks”. Without it the two orientations of one pattern pair may disagree, and the birth flip (round-18’s vindicated clause) is unsound.
<code>f_reads_rows</code>	The steering function is a function <i>of the rows</i> : it reads only in-range core and slot columns. Without it, steering could distinguish the frame’s padding from template junk outside the interface, and the recursive row reconstruction of U3 would not determine the recorded value.

This is the round-19 thesis operating at the next level: in prose, each of these would have been a referee’s defect #13; in Lean, they were unprovable-goal errors during round-20’s development, fixed before any reviewer saw them.

3 Provenance and hygiene

`doStep` was refactored for provability: the `let`-bound pattern and steering blocks became named definitions (`memberPortsList`, `patternVals`, `steeringVals`) and the Boolean pattern guard became propositional. The fold is extensionally the round-19 fold: the kernel-checked witness theorems (`defect12_written_gap`, the repair coverage and frame-agreement lemmas) re-verify unchanged, and WP34 Part B re-checks the Python transcription against it on 14,338 pairs with zero mismatches — now a *regression against a proved theorem* guarding the Lean/Python seam, no longer the only evidence. WP34’s Part-A expectation changed from “exactly one `sorry`” to “zero”; its Part-B steering function now reads only in-range row entries, the literal form of `f_reads_rows`.

4 What remains

The transport layer is now closed: definitions and their characterization are both in the kernel. Above it, the round-16 S-conditions, the canonical completion, and the inclusion theorem remain prose-plus-harness (their Lean port is the natural round-21); below it, the standing completeness mathematics (F6 width budgets, $W2'$ uniformization) and the single external input (the RCC5 patchwork property) are untouched. Ledger: thirteen reviews — twelve defects with witnesses, one no-counterexample; rounds 19–20 are unreviewed, and their natural review remains: attack the Lean statements. The decidability theorem for $\mathcal{ALCT}_{\text{RCC5}}$ remains **strongly supported, not certified**.